

Prüfer Code and Its Uniqueness for Trees

Mohammad Hossein Kazemini - Mohammad Javad Abdolahi

2025-02-25

1 Definition of Prüfer Code

The **Prüfer code** is a numerical sequence assigned to every labeled tree with n vertices. It provides a compact representation of the tree and establishes a one-to-one correspondence between labeled trees and sequences of length $n - 2$.

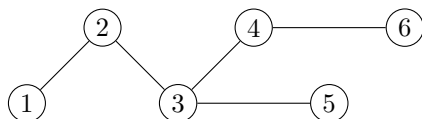
2 Algorithm to Generate Prüfer Code from a Tree

To generate the Prüfer code of a tree with n vertices:

1. Find the smallest leaf (a vertex of degree 1).
2. Record its neighbor in the Prüfer sequence.
3. Remove the leaf from the tree.
4. Repeat until only two vertices remain.

2.1 Example

Consider the following tree:



The process of generating the Prüfer code:

Step	Removed leaf	Prüfer code so far
1	1 (connected to 2)	2
2	2 (connected to 3)	3
3	5 (connected to 3)	3
4	3 (connected to 4)	4

Thus the Prüfer code of this tree is:

$(2, 3, 3, 4)$

3 Algorithm to Reconstruct the Tree from a Prüfer Code

To reconstruct the tree from a Prüfer code:

1. Compute the degree of each vertex.
2. Find the smallest vertex with degree 1 and connect it to the first element of the Prüfer sequence.
3. Decrease the degree of the connected vertex, remove the leaf, and continue.
4. When only two vertices remain, connect them.

3.1 Example

Using the Prüfer code $(2, 3, 3, 4)$ we reconstruct the tree:

Step	Edge added	Remaining Prüfer code
1	$1 - 2$	$(3, 3, 4)$
2	$2 - 3$	$(3, 4)$
3	$5 - 3$	(4)
4	$3 - 4$	$()$

The original tree is reconstructed (same shape as above).

4 Proof of Uniqueness

Theorem 1. *Every tree has a unique Prüfer code.*

Proof. The encoding process always removes the smallest leaf and records its unique neighbor. Since the smallest leaf is uniquely determined at each step, two different trees cannot produce the same Prüfer code. $\square$

Theorem 2. *Every Prüfer code corresponds to a unique tree.*

Proof. In the reconstruction process, at each step the smallest vertex of degree 1 is uniquely determined and connected to the first element of the remaining code. This deterministic process yields a unique tree. $\square$

5 Conclusion

The Prüfer code not only provides a compact representation of a tree but also allows us to count the number of labeled trees. The number of labeled trees with n vertices is:

$$n^{n-2}$$

which equals the number of distinct Prüfer sequences.